

SUMMERVILLE, SC, USA

WMO: 720632

Lat: 33.063N

Lon: 80.279W

Elev: 17

StdP: 101.12

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00218

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	-2.0	-11.2	1.4	0.7	-7.9	1.9	2.5	8.1	13.4	7.2	9.4	1.1	10	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	35.2	25.5	33.7	25.0	32.6	24.7	27.1	31.8	26.5	30.7	25.9	30.0	2.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	28.6	25.2	20.4	27.7	25.0	20.1	27.6	84.6	32.1	82.4	30.7	80.4	29.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.8	5.6	4.8	-6.6	36.8	3.0	1.2	-8.7	37.6	-10.4	38.3	-12.1	39.0	-14.2	39.9		
			-7.4	28.0	2.6	0.7	-9.2	28.5	-10.7	28.9	-12.2	29.3	-14.0	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	8.3	11.5	13.5	18.2	22.4	26.0	27.1	26.8	24.5	18.9	12.8	10.9
	DBStd	7.65	5.29	5.19	4.76	3.81	3.03	2.25	1.65	1.66	2.39	4.25	4.45	5.27
	HDD10.0	228	94	41	21	1	0	0	0	0	0	1	19	53
	HDD18.3	1186	312	197	163	49	7	0	0	0	1	46	175	237
	CDD10.0	3309	43	83	128	245	384	481	530	520	436	276	102	82
	CDD18.3	1226	2	6	12	44	133	231	272	262	187	63	8	7
	CDH23.3	10856	8	51	121	423	1160	2182	2566	2347	1444	482	49	23
CDH26.7	4412	0	3	19	94	410	1011	1182	1024	552	113	3	1	
Wind	WSAvg	1.6	1.9	2.1	2.1	1.9	1.5	1.3	1.2	1.1	1.3	1.4	1.6	1.8
Precipitation	PrecAvg	1193	85	91	93	77	86	131	135	149	104	89	67	86
	PrecMax	1472	184	209	194	159	173	251	277	276	253	337	186	186
	PrecMin	874	34	36	12	23	29	64	66	82	7	0	4	21
	PrecStd	176	45	42	40	34	37	47	52	48	51	75	45	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	27.2	29.0	31.1	35.1	37.1	37.1	35.9	33.7	31.3	27.3	26.0
		MCWB	18.8	18.8	18.5	21.3	23.1	25.8	25.8	26.1	24.6	23.9	21.2	21.3
	2%	DB	21.5	24.9	27.1	29.1	32.4	35.0	35.2	34.0	32.4	29.8	25.1	23.0
		MCWB	17.2	18.1	18.4	20.4	22.0	25.1	25.9	25.5	24.3	23.4	19.6	19.6
	5%	DB	19.8	22.5	24.8	27.5	31.0	32.9	33.0	32.7	31.3	27.6	22.7	22.1
		MCWB	15.9	16.6	16.8	19.6	22.2	24.2	25.5	25.2	23.9	21.6	18.3	19.5
	10%	DB	17.5	20.3	22.4	26.1	29.1	32.3	32.4	32.2	30.1	26.2	21.2	20.0
		MCWB	14.8	16.0	16.2	18.7	21.9	24.1	25.3	25.1	23.5	20.9	18.0	18.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	20.7	21.0	23.4	24.5	27.4	28.2	27.5	26.4	25.5	22.8	22.7
		MCDB	21.9	23.7	24.3	27.2	29.4	32.5	34.3	31.8	29.4	27.9	25.1	24.4
	2%	WB	18.9	19.4	19.8	22.4	23.8	26.4	27.2	26.9	25.7	24.5	21.3	21.4
		MCDB	21.6	23.3	24.0	26.1	28.7	31.8	32.3	31.0	29.0	27.8	22.7	22.5
	5%	WB	17.4	18.1	18.8	21.4	23.2	25.6	26.5	26.3	25.2	23.5	20.2	20.0
		MCDB	18.1	21.3	23.1	24.9	27.9	30.8	31.1	30.2	28.7	26.5	21.9	21.1
	10%	WB	15.2	16.8	17.5	20.5	22.7	25.0	25.8	25.8	24.6	22.6	18.5	17.8
		MCDB	16.8	19.1	20.2	23.9	27.2	30.1	30.2	29.5	28.4	25.0	21.1	19.0
Mean Daily Temperature Range	5% DB	MDBR	12.0	12.4	13.2	13.5	12.0	11.4	10.7	10.2	10.5	12.0	12.8	11.2
		MCDBR	14.3	14.4	15.3	15.0	14.8	13.1	12.0	11.5	12.0	12.5	13.8	12.4
		MCWBR	9.3	8.0	7.3	6.8	5.6	4.3	4.4	4.2	4.6	6.3	8.7	8.2
	5% WB	MCDBR	11.0	12.5	12.0	11.5	11.3	11.5	11.1	10.4	9.8	10.1	10.3	10.2
		MCWBR	8.9	8.1	6.4	6.0	4.8	4.3	4.5	4.2	4.2	5.4	7.6	7.5
Clear-Sky Solar Irradiance	taub	0.324	0.333	0.366	0.408	0.471	0.507	0.544	0.520	0.437	0.392	0.346	0.328	
	taud	2.527	2.497	2.423	2.293	2.160	2.111	2.023	2.085	2.312	2.411	2.500	2.548	
	Ebn at Noon	895	922	911	882	826	793	761	773	829	843	860	870	
	Edh at Noon	84	95	111	132	152	159	173	160	123	103	86	79	
All-Sky Solar Radiation	RadAvg	2.86	3.50	4.61	5.86	6.30	6.43	6.19	5.59	4.77	4.10	3.17	2.54	
	RadStd	0.15	0.36	0.40	0.39	0.45	0.45	0.39	0.33	0.40	0.46	0.23	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+206	+144

Nomenclature: See separate page